RUTH ELVIN HARIANJA

DATA SCIENTIST | AI & MACHINE LEARNING

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PROFESSIONAL SUMMARY

Data Scientist & Master's graduate in Data Science & Artificial Intelligence with a strong portfolio in end-to-end machine learning, predictive modeling, time series forecasting, and Explainable AI (XAI). Proficient in Python (Pandas, NumPy, Scikit-Learn, TensorFlow/Keras, PyOD, Statsmodels). Proven track record of developing production-grade ML web applications (Flask, Streamlit), spatio-temporal anomaly detection engines (Isolation Forest, LOF, ECOD, DBSCAN, ST-DBSCAN), time series forecasting models (LSTM vs ARIMA), and complex behavioral feature engineering pipelines on large-scale datasets.

TECHNICAL SKILLS & COMPETENCIES

• **Machine Learning & AI:** Scikit-Learn, TensorFlow/Keras, PyOD (Isolation Forest, LOF, ECOD), Random Forest, Logistic Regression, SHAP (XAI), Genetic Algorithm, Perceptron ANN.

• **Data Science & Analytics:** Python (Pandas, NumPy), SQL, Time Series Forecasting (LSTM, ARIMA), Advanced Feature Engineering, Exploratory Data Analysis (EDA), Statsmodels.

• **Geospatial & Web Deployment:** Spatial DBSCAN, ST-DBSCAN (Spatio-Temporal), Haversine Distance, Dynamic Centroid Estimation, Flask, Streamlit, PyDeck 3D, Leaflet.js, Folium, Plotly.

FEATURED DATA SCIENCE & AI PROJECTS

Attendance Anomaly Detection *(Flask, PyOD, SHAP, Leaflet.js)* **2026**

- Built an end-to-end production ML web app deploying Isolation Forest, LOF, and ECOD ensemble voting for geospatial anomaly detection.
- Integrated SHAP XAI for root-cause feature attribution, automated preprocessing pipelines, and interactive Leaflet.js mapping.

Geospatial & Spatio-Temporal Anomaly Scoring Dashboard *(Streamlit, ST-DBSCAN, PyDeck 3D)* **2026**

- Developed an interactive 3D analytics dashboard implementing ST-DBSCAN and Haversine distance for spatial/temporal noise identification.
- Engineered a multi-factor weighted risk scoring algorithm categorizing anomaly risk levels (LOW, MEDIUM, HIGH) for executive decision-making.

Beverages Sales Time Series Forecasting Engine *(TensorFlow LSTM vs ARIMA)* **2025**

- Modeled 55k+ retail transactions comparing ARIMA baseline with LSTM Neural Networks to forecast volume and capture non-linear trends.
- Integrated promotional exogenous drivers (onpromotion) and commodity prices, proving promotional lift impact over macroeconomic drivers.

Credit Card Fraud Detection & Behavioral Pipeline *(Scikit-Learn, Random Forest, Imbalanced Learning)* **2025**

- Constructed a high-precision fraud detection engine on extreme imbalanced data (0.58% fraud rate) with class_weight='balanced'.
- Engineered complex behavioral features: transaction velocity (velocity_1h), time delta (diff_trans), rolling averages, and Haversine distance.

Metaheuristic Optimization & Neural Algorithm Development *(Python, NumPy from scratch)* **2024**

- Designed a custom Genetic Algorithm (SchedulingGA) for multi-constraint resource optimization and built a Perceptron ANN from scratch.

WORK EXPERIENCE

Quality Assurance Engineer — Gokomodo | *Full-time (South Jakarta - On-site)* **Oct 2022 – Aug 2024**

- Executed test scenarios, test cases, Agile cross-functional collaboration, and backend database validation for B2B e-commerce platforms.

Software Quality Assurance — Anteraja | *Contract (South Jakarta - Hybrid)* **Oct 2021 – Sep 2022**

- Conducted functional, integration, and regression testing for high-volume logistics and tracking systems using JIRA and TestRail.

Software Quality Assurance Intern — Transfree | *Internship (South Jakarta)* **May 2021 – Jul 2021**

- Assisted in creating test plans, smoke testing, and UAT for cross-border financial transaction mobile applications.

EDUCATION

Magister Sains Data dan Kecerdasan Buatan (S2 DSAI) | *Universitas Sumatera Utara (USU)* **2024 – 2026**

D-IV Teknologi Informasi / Rekayasa Perangkat Lunak | *Institut Teknologi Del* **2017 – 2021**